

School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing

Lecture-11 Scribe Submission

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CSE400 – Lecture 11 Scribe

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1 Lecture Structure and Objective

1.1 Outline of the Lecture

The lecture consisted of three main components:

1. Transformation of Random Variables
 - Learning transformation techniques for random variables.
2. Function of Two Random Variables
 - Understanding joint transformations and derived distributions.
3. Illustrative Example
 - Detailed derivation for the case:

$$Z = X + Y$$

2 Transformation of Random Variables

2.1 Definition and Notation

Let X denote an original random variable.

Let Z denote a new random variable obtained through transformation of X .

Thus, a transformation is represented as:

$$Z = g(X)$$

2.2 Objective of Transformation

Transformation techniques are used to:

- Derive new random variables from existing ones.
- Study how the properties of the original variable affect the transformed variable.

2.3 Reasoning and Logical Development

Step-wise reasoning presented in lecture:

1. Begin with a known random variable.
2. Apply a mathematical function to the variable.
3. The output of the function becomes a new random variable.
4. Transformation techniques help determine the behavior and characteristics of the new variable.

2.4 Assumptions and Conditions

- The transformation is defined using a function applied to an existing random variable.
- The resulting value is treated as a random variable.

3 Function of Two Random Variables

3.1 Definition and Notation

Let X and Y denote two random variables.

Let Z denote a new random variable obtained from both variables.

The transformation is represented as:

$$Z = g(X, Y)$$

3.2 Concept of Joint Transformation

Joint transformation involves:

- Combining two random variables.
- Producing a new random variable using a function of both variables.
- Studying derived distributions based on the joint behavior of the original variables.

3.3 Logical Step-by-Step Reasoning

1. Start with two random variables.
2. Define a function involving both variables.
3. Apply the function to generate a new random variable.
4. Analyze the derived distribution of the new variable.

3.4 Result Emphasized in Lecture

- Functions of multiple random variables lead to derived distributions.
- Joint transformation extends the idea of transformation from one variable to multiple variables.

4 Illustrative Example: $Z = X + Y$

4.1 Statement of the Example

The lecture provided a detailed derivation example using:

$$Z = X + Y$$

4.2 Objective of the Example

- To demonstrate practical application of joint transformation.
- To illustrate how a new random variable is formed from two variables.

4.3 Step-by-Step Reasoning Provided

1. Consider two random variables X and Y .
2. Define a new variable Z as the sum of the two variables.
3. The value of Z depends on the combined values of X and Y .
4. The derived distribution of Z is determined from the joint behavior of X and Y .

4.4 Key Insight from Example

- Summation of two random variables is a special case of transformation involving two variables.
- The example demonstrates how transformation techniques are applied to derive new distributions.

5 Logical Flow of Ideas Across the Lecture

The lecture followed the progression:

1. Introduction to transformation of a single random variable.
2. Extension of transformation techniques to functions involving two random variables.
3. Application of concepts using the summation example.

6 Key Takeaways

- Transformation methods allow creation and analysis of new random variables.
- Joint transformations consider relationships between multiple variables.
- Derived distributions describe outcomes resulting from transformations.
- The example $Z = X + Y$ demonstrates practical implementation of transformation concepts.